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PUBLICATIONS

- H. Kumar, **A. Konkar**. *Simple Transformer with Single Leaky Neuron for Event Vision*. **Feb 2025**
Proceedings of the Winter Conference on Applications of Computer Vision (WACV) Workshops, 2025, pp. 928-934
- A. Konkar**, X. Qu. *A Review of Transformer-Based and Hybrid Deep Learning Approaches for EEG Analysis*. **Jun 2025**
International Conference on Human-Computer Interaction (HCI International) 2025.

EDUCATION

- Master of Science in Computer Science** **May 2025**
The George Washington University *GPA: 3.71/4.0*
Thesis: Enhancing EEG-Based Gaze Prediction with Transformers on EEGEyeNet
Relevant Courses: Computational Linear Algebra, Machine Learning, Neural Networks & Deep Learning, Computer Vision
- Bachelor of Engineering in Information Technology** **Oct 2020**
University of Mumbai

EXPERIENCE

- Research Assistant – Computer Vision** **Oct 2025 – Present**
GW Vision Lab *Washington, DC*
- Creating a dataset of event camera recordings of various objects responding to sound stimuli, where audio not recorded.
 - Since event cameras can capture tiny, rapid brightness changes, sound becomes visible in the form of motion as it physically vibrates objects. Checkout this post to view more information: [See Motion from Sound with Event Cameras](#).
 - Investigating methods to reconstruct acoustic signals from event-based visual input. Research Advisor: Dr. Robert Pless.

- Software Engineer** **Mar 2025 – Present**
National Collegiate Table Tennis Association (NCTTA) *Remote*
- Developing and shipping production-ready features using ASP.NET Core for the NCTTA web application serving **20,000+** users (see live app: <https://nctta.app/>). This app manages all U.S. collegiate-level table tennis tournaments.
 - Improved service performance and scalability by **20%** through SQL query optimization, indexing strategies, and streamlining request-response data flows to reduce backend latency and server load.

- Research Assistant – Data Science** **Nov 2023 – Oct 2025**
GW Institute of Public Policy *Washington, DC*
- Reanalyzed large multi-site randomized controlled trials (*HPOG 1.0* and *PACE/Year Up*; 23,000 participants, 50+ sites) to estimate site-specific and out-of-sample program impacts beyond the original evaluations.
 - Built data pipelines in R to integrate multi-wave survey data, job-spell records, and site-level program and labor-market variables from restricted-use *ICPSR* files, handling complex joins, cohort overlap, and data-quality constraints.
 - Constructed primary and secondary outcome measures (e.g., weekly earnings, credential attainment) by transforming raw survey responses into consistent, comparable quantities across programs and time horizons.
 - Addressed missing covariate data using Multiple Imputation by Chained Equations (MICE), imputing separately by study population to preserve validity in downstream causal analyses.
 - Implemented hierarchical causal models with regularization and shrinkage (FIRC, LASSO, Empirical Bayes) to estimate heterogeneous treatment effects and improve out-of-sample reliability.
 - Evaluated model generalization using out-of-sample projections and uncertainty estimates to assess how well impacts transport across sites and settings.
 - Impact: Provided a framework for site-specific impact estimation with quantified uncertainty, aimed at informing adoption decisions beyond average effects.

- Software Engineer** **Aug 2020 – Jun 2022**
Larsen & Toubro Infotech *Mumbai, India*
- Developed & maintained microservices with Spring Boot for a large-scale capital markets platform for our client, *Citi*, ensuring compliance with corporate QA standards. Tech Stack: Java, Spring Boot, Selenium, TestNG.
 - **Saved 8 hours** of manual testing effort **per week** by automating complex end-to-end test scenarios.
 - Collaborated with cross-functional teams to translate functional specifications into modular, maintainable software components. Participated in code reviews and iterative feature delivery. Followed Agile methodologies and TDD.
 - Deployed and operated containerized microservices on AWS EC2 using OpenShift, gaining hands-on experience with cloud-native deployment, scaling, and service orchestration.

TEACHING EXPERIENCE

Teaching Assistant – CSCI 1011. Introduction to Software Development, GW

May 2025 – Aug 2025

Teaching Assistant – CSCI 1112. Algorithms and Data Structures, GW

May 2025 – Aug 2025

Teaching Assistant – CSCI 2113. Software Engineering, GW

May 2025 – Aug 2025

Student Tutor – *GW Athletics*

Sep 2023 – May 2025

- MATH 1221. Calculus with Precalculus II.
- MATH 1232. Single-Variable Calculus II.
- MATH 3125. Linear Algebra II.
- CSCI 1011. Introduction to Programming with Java.
- CSCI 1112. Algorithms and Data Structures.

SELECTED PROJECTS

FlavorSignal – Serpapi, Ollama(Phi-3 Mini), Next.js, Python

[GitHub](#)

- **Outcome:** Given your location and favorite dish as input, FlavorSignal fetches all the nearby restaurants and tells you whether that dish is worth ordering by cutting through noisy reviews and summarizing opinions specific to your dish.
- **Approach:** FlavorSignal filters reviews semantically at the dish level, using query expansion and embedding-based retrieval to isolate only text relevant to the requested item.
- The filtered, item-specific reviews are summarized to provide a concise assessment of whether a dish is worth ordering, including sentiment trends and common complaints.

From-Scratch Implementation of a Low-Level Image Classification Network – Python

[\[Report\]](#) [GitHub](#)

- Developed a complete neural network training pipeline in Python (without using deep learning libraries).
- Implemented manual forward pass, backpropagation, gradient updates, and weight initialization.
- Built custom image preprocessing modules (posterization, enhancement, feature extraction) to improve data quality and model performance.

Landmark Recognition – Python, Streamlit

[GitHub](#)

- Developed a landmark recognition web application that predicts landmarks from images, retrieves their full address with latitude/longitude, and visualizes them on an interactive map for exploration.
- Leveraged a pretrained tensorflow-hub model, trained on the Google Landmarks Dataset V2.

Real-Time Person Detection & Tracking – Python

[GitHub](#)

- Developed a real-time person detection and tracking pipeline using YOLOv8x for detection and DeepSORT for tracking.
- Evaluated multiple tracking methods (IOU, SORT, DeepSORT). Tested on an NVIDIA RTX 3070.

Forecasting Hourly Electricity Demand and Assessing Grid Resilience – Python

[\[Report\]](#)

- Fine-tuned machine learning models (LSTM and Prophet) to predict hourly electricity demand from national grid data.
- Evaluated how major disruptions (like storms or accidents) influence grid performance, turning data-driven insights into recommendations for stronger energy systems.

TECHNICAL SKILLS

Programming Languages: C, C++, Embedded C, Python, MATLAB, R, SQL, Java, JavaScript

Frameworks & Libraries: PyTorch, Keras, Tensorflow, NumPy, Pandas, OpenCV, Scikit-learn, Matplotlib

DL Architectures: MLP, CNN, RNN, LSTM, Self-Attention, Transformers, Attention-Based Fusion, VAE, GAN, LLM

Computer Vision: Image formation & camera optics, Camera calibration & geometry, Feature tracking, Optical flow, Pose estimation, SfM, Visual Odometry, SLAM

Domain Skills: Event-Based Vision, Spiking Neural Networks, Robot control, State estimation, Sensor fusion, Motion planning, ROS, CUDA, ONNX, TensorRT, Linux, BASH

REFERENCES

Robert Pless

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